

Hydrology modelling of Rajkot District using QGIS

Internship Report
Arvind Ltd.

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Acknowledgment

The internship opportunity I had with Arvind Ltd. was a great chance for learning and professional development. And, I consider myself honored. I am also grateful for having a chance to meet so many wonderful people and professionals who were very kind and welcoming.

I would like to give my special thanks to Prof. Ratna Ghosal who believed in my potential and recommended me for an internship at this highly esteemed organization, Arvind Ltd.

I am using this opportunity to express my deepest gratitude and special thanks to Mr. Punit Lalbhai, Executive Director, and Mr. Abhishek Bansal, Head of Sustainability, who in spite of being extraordinarily busy with their duties, took time out to consider my application for this internship and provided me with a chance.

I express my deepest thanks to Mr. Snehit Roy Kumar, Senior Manager for always attending my queries, helping me fix the errors that I would face during the project, giving me necessary advices and guidance which were extremely valuable for my study both theoretically and practically.

I would also like to thank HR Head, Ms. Bhavana Ojha who made sure my experience at Arvind Ltd is smooth and was always a call away.

I perceive this opportunity as my first and very significant step towards my career development. I will strive to use gained skills and knowledge in the best possible way, and I will continue to work on their improvement, in order to attain desired goals.

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Introduction

Watershed and its Significance:

Watershed is a geo-hydrological unit, the area where all the water that enters comes out of a common outlet or the main waterbody. Productive status of a watershed is affected by the natural and human activities, type of land, soil, people, and livestock. Thus, it's known to be a biological, physical, economic and social system. An efficient management of watershed areas is considered as the best solution for the rising food, and water scarcity in many parts of the world.

Scenario in India:

The hot fragile climate of India and the high variability in rainfall patterns along with low investment in resource management practice calls for active programmes that are focused towards management of land surface and vegetation so as to conserve and utilize maximum water that falls on the area of watershed, moderate floods, mitigate drought, decrease the soil erosion, and reduce silting up, which shall cumulatively benefit the farmer enhancing the crop yield.

Various watershed programmes are implemented across the country. One of them is Integrated Watershed Management Programme (IWMP) implemented by Department of Land Resources of Ministry of Rural Development. Their objective is to restore ecological balance by harnessing, conserving and developing degraded natural resources such as soil, vegetative cover and water.

In 2018, National Institute of Transforming India (NITI Aayog) launched a composite water management index and ranked all the states accordingly. Gujarat among them has retained rank one.

Role of GIS:

The use of satellite images and working with GIS provides a better understanding of the situation. Through modelling studies in GIS tools (QGIS, ArcGIS), future predictions could be done and consequences could be known under different scenarios. This helps for better implementation of the idea on site.

This project involves studying the hydrology of the area of interest, the knowledge of which can be further used on field for implying conservation practices and/or building watershed conservation structures.

Objective

To create a hydrological model of villages from where the country source cotton through GIS.

- Download and import a Digital elevation model and generate channels and streams of the area of interest with the help of QGIS tools.
- Carrying out SWAT modelling of watershed hydrology based on the land use/land cover and soil map, and rainfall data.

Description of work/Methodology:

Procure Data:

- **DEM:** [click here](#)
Open archive data → select sub-category: Cartosat-1 → select product: CartoDEM all versions → In select area section → Tiles → On the map: while pressing Ctrl key select multiple tiles that covers your area of interest (here Gujarat)

- **Download an India Administration file**
- **LULC:** [click here](#)
Click on 'View' on top right side → select the boxes that is under your area of interest → download 2021 data (tiff file)
- **Soil map:** [click here](#)
Type 'digital soil map of the world' in the search box → click on that option → download 'Digital Soil Map of the World - ESRI shape file format'
*Direct download link: [click here](#)
- **Rainfall data:** SWAT global weather data and weather generator database were used in SWAT+ editor

Import data in QGIS:

- **Preparing Digital Elevation Model:**
Add DEM → Fill sinks → Merge DEM → Clip DEM according to the area of interest → Reproject the DEM to 43N UTM projection system.
- **Extracting our area of interest:**
Clip India administrative file to the area of interests (here BCI villages).
- Make sure that the **saga plugin is installed**
- From processing toolbox → under **terrain analysis** of saga plugin → select channel network → save the shape file.
- **Clip the Land use map** to the area of interest
Define the classes of a Land use map and assign names like following:
 1. WATR
 2. FRST
 3. AGRL
 4. URBN
 5. RNGE
- Clip the soil map and classify as given in the link: [click here](#)
*Rajkot district has two kinds of soils:
 1. Vc (Chromic Vertisols)
 2. I (Lithosols)

Collected literature studies to find the best suitable site to download the Land, soil and rainfall data for our area of interest. Once they were downloaded and further imported in QGIS, display changes were done from 'Symbolology' option to better understand and interpret the map. Classification was done from the same option, under the title 'Label' (refer Pg. 7 and 8 for Map 2 and 3 respectively).

3. SWAT model:

1. Delineate Watershed

- a. Select DEM – Input reprojected DEM
- b. Delineate Watershed option – Click on Burn in stream network and use the Channel Network Shape File
- c. Create Streams
- d. Don't Click on Use an Inlet / Outlets Shapefile – directly create watershed
- e. In Create Landscape – Click on it – Then use the DEM Inversion Method – this creates a flood plain
- f. In Merge Subbasins – Click on Select Sub basins – then click on Select small subbasins – and then use a mean area threshold
- g. Click on ok when the above steps are done.

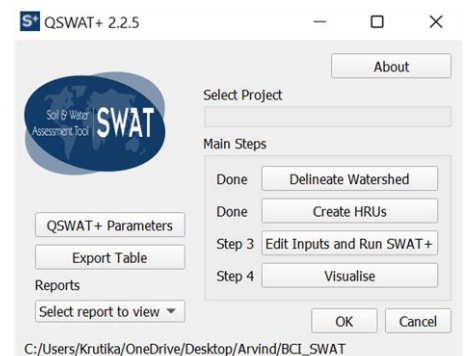


Fig 1: The dialog box appears when you click on S+ plugin.

2. Create HRUs

- a. Landuse and Soil tab
 - i. Select the land use map and Soil Map from Input Data
 - ii. In Tables
 1. Landuse Lookup – Use the LULC CSV file
 2. Soil Lookup – Select Global_souls
 3. Usersoil - Select global_usersoil
 4. Plant – Select plant
 5. Urban – Select Urban
 - iii. In Set Slop Bands – Write 10 and click on Insert
 - iv. In Soil data – select usersoil
 - v. In Select Flood plain map – select the flood plain map crated in Create Landscape option (Point 1 (e))
 - vi. Click on Generate Full HRUs
 - vii. Click on Read
- b. HRUs
 - i. In Single/Multiple HRU's – Select Dominant HRU and then click on Create HRUs.

3. Click on Edit Inputs and Run SWAT +

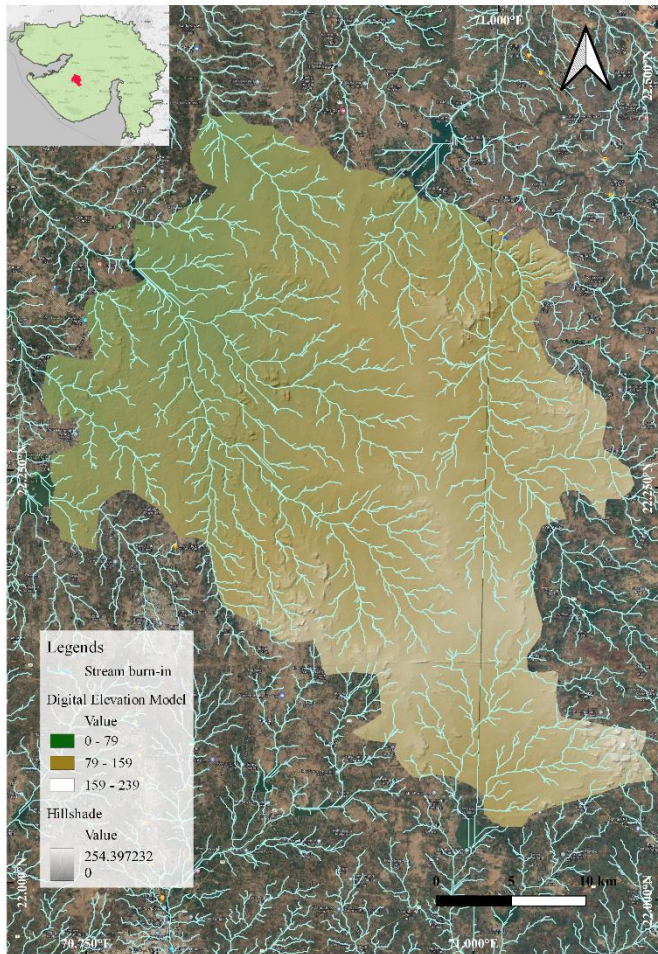
- a. Open the project of interest (here 'BCI SWAT')
- b. On left side panel, open Edit SWAT+ inputs → import default global weather generator and weather station data provided by SWAT through import data option in each of them
- c. On left side panel, click on run SWAT+ → under confirmation stimulation settings set your simulation time period
- d. Under choose output to print, select all the checkboxes under monthly as well → click on save settings and run selected.

4. Visualisation:

- a. From QSWAT 2.2.5 dialog box, select step 4 'visualise' → choose the output table, select the time period → click on plot → select unit and variable → add plot → click on plot → locate and name the file → analyse the excel files.

Observations/Results:

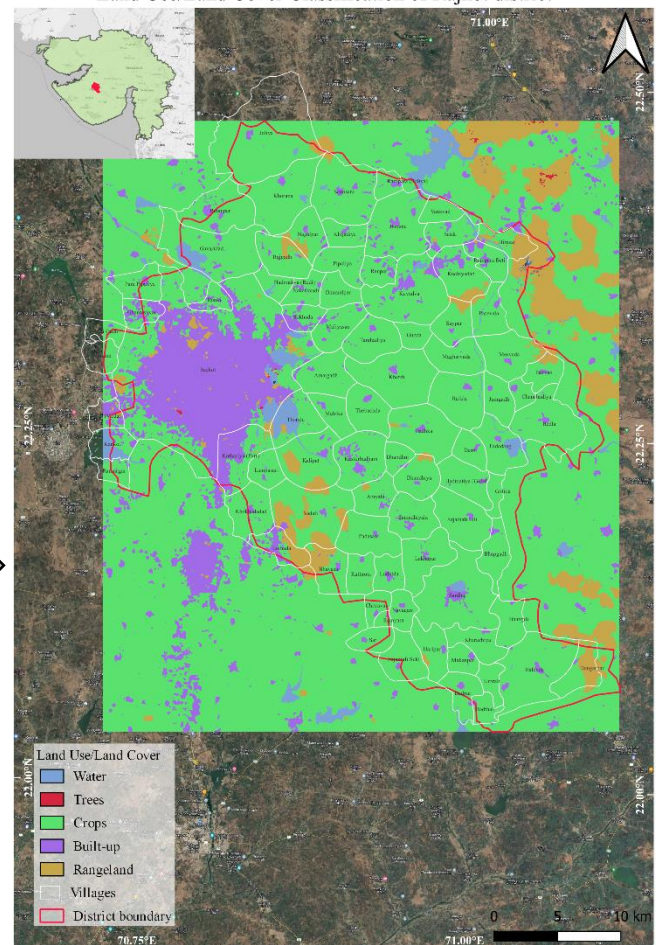
Streams of Rajkot District and Surrounding Areas



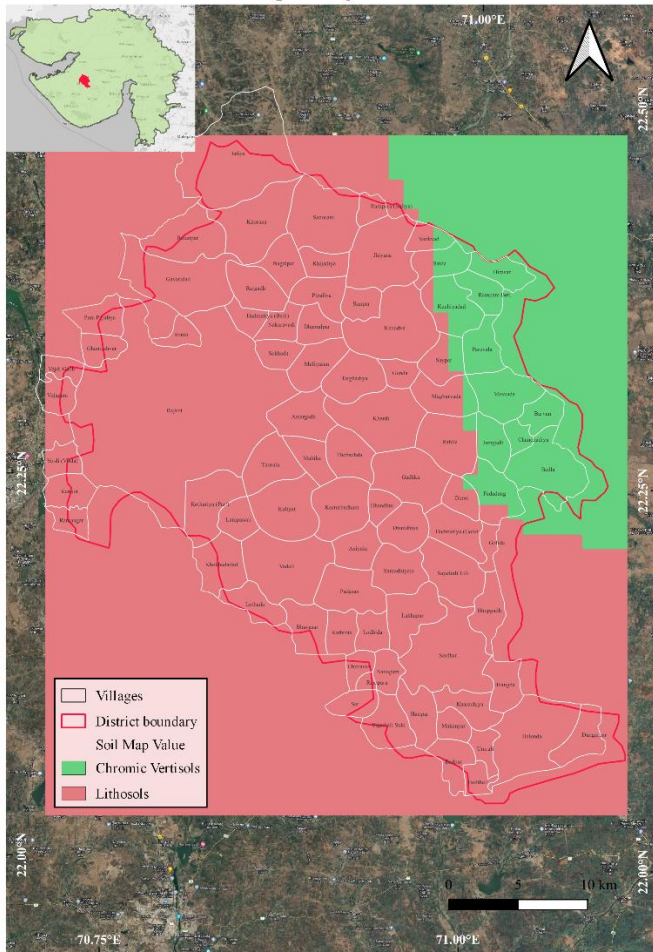
Map 1:
Streams generated through SWAT+ model.
Saga terrain analysis could also be used
for similar result.

Map 2:
Land use map with the Rajkot district demarcation
showing higher coverage of agricultural or crop
land followed by built-up area.

Land Use/Land Cover Classification of Rajkot district



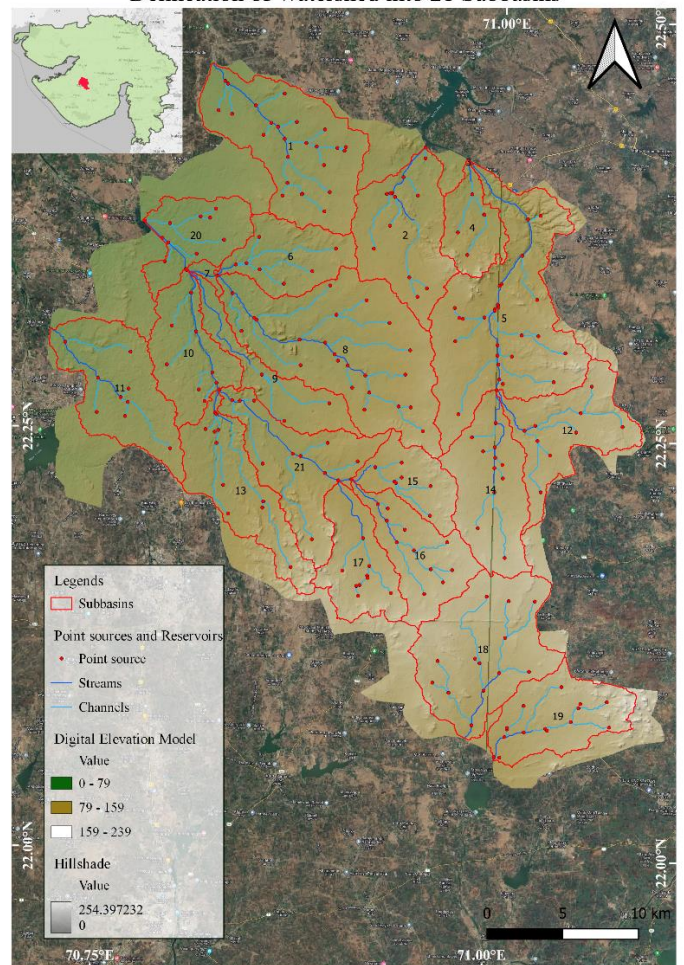
Soil Map of Rajkot District



Map 3:

Soil map with Rajkot district demarcation showing two kinds of soil types in the villages of Rajkot. Vertisols are dark-coloured cracking and swelling clays, while Lithosols are a group of shallow azonal soils consisting of imperfectly weathered rock fragments.

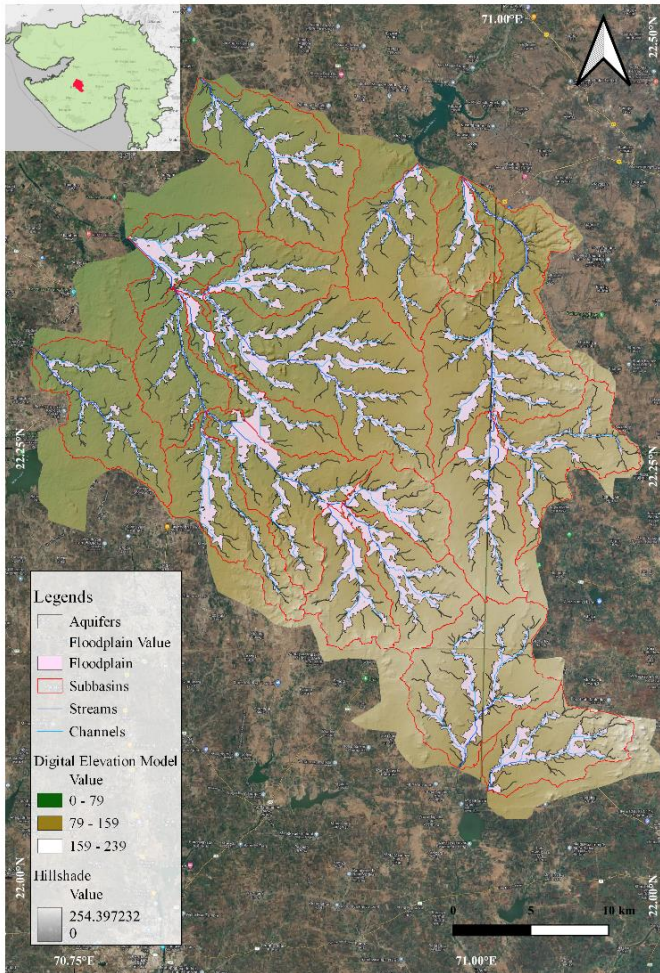
Delineation of Watershed into 21 Subbasins



Map 4:

Results obtained when 'Delineate watershed' option is selected; sub basins with respective streams and channels are generated.

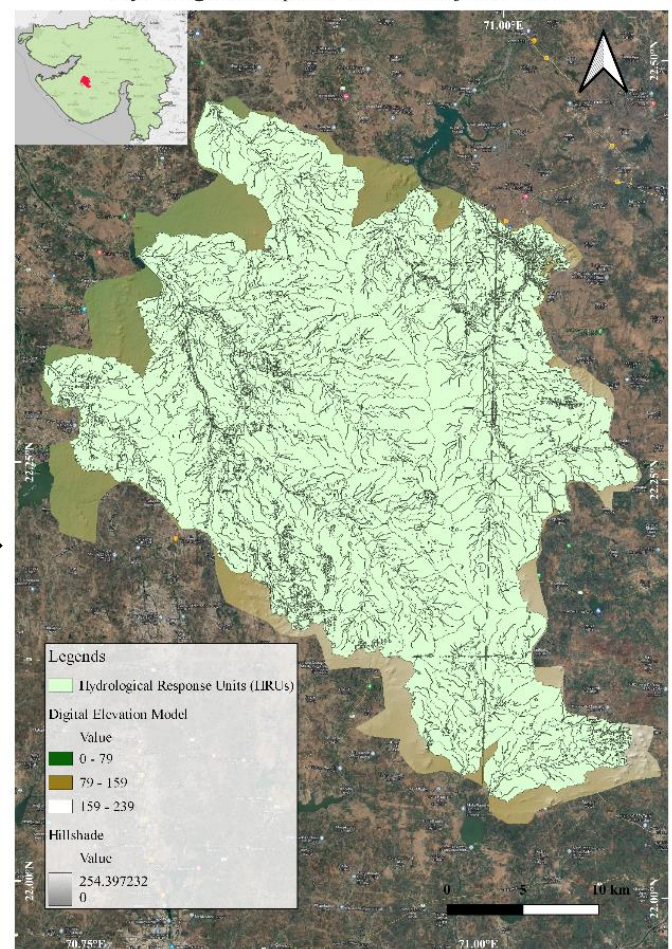
Aquifers and Floodplain Areas of Rajkot District



Map 5:

Aquifers and Floodplains area around the channels are generated when respective options are selected in 'delineate watershed' dialog

Hydrological Response Units of Rajkot District



Map 6:

After the watershed delineation, the 'create HRUs' option (Step 2) opens wherein the Land use map, soil map and rainfall data is

Further the HRU file extracted is linked beside [click here](#). The file summarises the above maps into data for all the 21 sub basins.

The final steps and results are as follows:

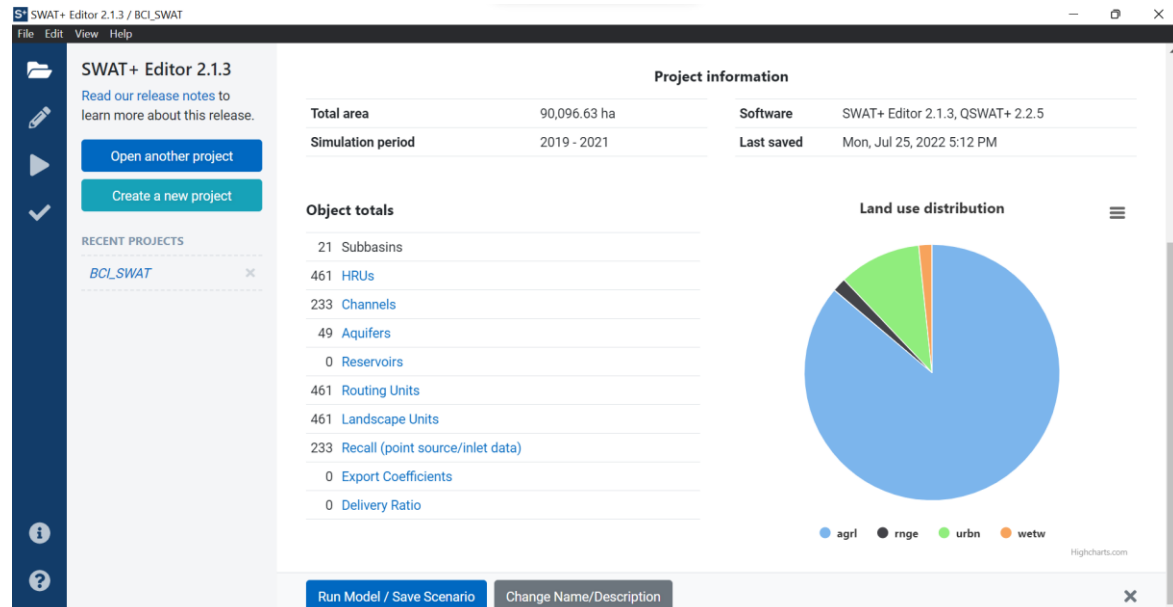


Fig 2: The above Editor opens once step 3 i.e. 'Edit inputs and run SWAT+' is processed. It is an informative page that projects the Land use distribution in the form of pie chart and the number of subbasins, HRUs, channels, etc.

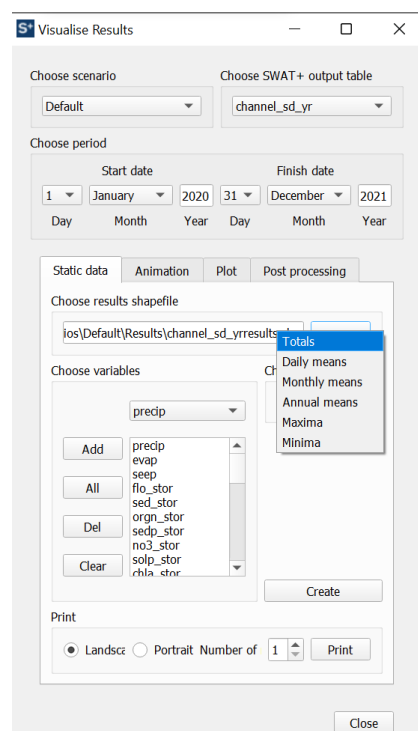


Fig 3: the step 4 is to 'visualize', the above dialog box appears and the fields are entered based on the interest of the project. 'Yearly channel water balance, incoming/outgoing sediment and nutrient and storage' was selected in the above image, all variables were selected, with total values. Click on 'create' option to download and analyse the result table.

Discussion and Future work

Above maps were created using QGIS features and associated tools/plugins. In Map 2 it is observed that the agricultural land has the highest coverage thus it calls for efficient and sustainable management. Moreover, the built area is concentrated on the top Left side and hence the impact of it on the watershed area and flow direction could be understood in further studies. Based on the soil map created (refer Map 3), appropriate practices could be implemented which is beneficial to the area of interest with a particular soil type.

The SWAT+ editor has the features to add various construction structures such as trenches, filter strips, bounds and further to see the results- change in run-off, and find the best method for the target area. For this process Map 4 is helpful which provides a clear picture of streams and channels along with delineated sub basins.

Floodplain area are around the rivers and streams (as also observed in Map 5). It is the flat area of land that is likely to flood or in other words, land built up from soil left by the floods. They get sediment carrying floodwaters that enrich the soil and so are considered to be a good agricultural land. They provide the space for rivers to spread their waters which makes such areas really important to be considered and analysed during any kind of development near rivers, streams and likewise waterbodies. Floodplain edges are marked and named aquifers that indicates the extent to which the flood can reach.

At the end Hydrological Response Units (HRUs) are also created. They are the portions of a subbasin that possess unique landuse/management/soil attributes. In one subbasin, Landuse, soil type and such characteristics are scattered and hence HRU is an area that clubs the similar ones in a single response unit.

Conclusion:

Based on the current work done and to bring this project on field, a lot of interpretation and analysis is required from the numerous result tables that could be obtained under different scenarios. Considering the tight time window of the internship, I believe I was able to create a primary base for this study that could be further taken forward to produce results that could help make decisions around potential sustainable development in that area.

The problem of water scarcity and low crop yield has been faced in many villages across the country. Therefore, working with satellite data and doing the trial and error work on a computer rather than on field would have higher chance of success rate.

References:

[https://www.india.gov.in/integrated-watershed-management-programme-ministry-rural-development#:~:text=Integrated%20Watershed%20Management%20Programme%20\(IWMP,soil%20C%20vegetative%20cover%20and%20water.](https://www.india.gov.in/integrated-watershed-management-programme-ministry-rural-development#:~:text=Integrated%20Watershed%20Management%20Programme%20(IWMP,soil%20C%20vegetative%20cover%20and%20water.)

<https://www.soilmanagementindia.com/soil-erosion/watershed-development-and-management-in-india/3130>

<http://arkgroup.arkgroup.co.in/upload/Article/9441.%20GUJARAT-%20%20A%20%20LEADING%20STATE%20IN%20WATERSHED%20DEVELOPMENT-%20A%20REVIEW%20%20PATEL,V.R..pdf;%20%20PATEL,%20B.M.%20%20AND%20%20%20PAVAYA,%20R.P..pdf>

<https://www.indiawaterportal.org/articles/sad-state-floodplains>

https://www.researchgate.net/publication/256202770_Studies_on_Watershed_Management_in_Shivaji_University_Kolhapur_India

http://www.unishivaji.ac.in/uploads/journal/Journal_42/3.pdf

<https://www.ijrte.org/wp-content/uploads/papers/v8i2/B2275078219.pdf>

https://swat.tamu.edu/media/69302/ch01_overview.pdf

<https://swat.tamu.edu/media/99192/swat2009-theory.pdf>

<https://www.fao.org/3/aq361e/aq361e.pdf>